# EXPERIMENT 8

Configuring User Management and Role-Based Access Control (RBAC)
in Zoho Creator

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## Title

Design and Implementation of an Event Registration & Participant Management Application with User Permissions, Roles, Role Hierarchy and Data Sharing in Zoho Creator.

## Aim

To configure role-based access control in the Event Registration & Participant Management application by creating custom permission sets, defining a role hierarchy, and configuring a data-sharing rule in Zoho Creator.

## Objectives

• To configure custom permission sets for event organizers and registration staff.
• To establish the role hierarchy CEO → Event_Organizer → Registration_Staff.
• To configure record visibility using the Data Sharing tab and a Read Only access level.
• To understand how permissions, roles and data sharing work together to restrict access to application data.

## Software / Tools Required

• Zoho Creator web-based low-code application development platform
• Event Registration & Participant Management application
• Web browser with an active internet connection
• Zoho account with permission to edit the application

## Introduction / Theory

**User Management:** User management determines who can access an application and what each person can view or do. Zoho Creator groups these controls under Permissions, Roles and Data Sharing.

**Role-Based Access Control (RBAC):** RBAC assigns access rights through roles rather than configuring every individual separately. It supports least-privilege access and consistent administration.

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## Key Concepts

| Concept | Description |
|---|---|
| Permissions | Named permission sets define the actions a user can perform on application components and records. |
| Roles | Roles represent positions in the application's organizational hierarchy. |
| Role Hierarchy | A hierarchy establishes reporting relationships. Higher roles can gain visibility into records owned by subordinate roles, subject to permissions. |
| Data Sharing | A sharing rule extends record visibility from selected roles or users to other roles or users, with Read Only or Read/Write access. |

## Application Specifications

| Parameter | Detail |
|---|---|
| Application Name | Event Registration & Participant Management |
| Form Name | Event_Registration_Form |
| Custom Permission Sets | Organizer_Permission; Registration_Staff_Permission |
| Role Hierarchy | CEO → Event_Organizer → Registration_Staff |
| Data Sharing | Event_Registration_Form: Read Only access selected in the rule configuration screen |

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## Procedure

1. Sign in to Zoho Creator and open the Event Registration & Participant Management application in edit mode.
2. Select **Settings** → **User Permissions**. The Permissions, Roles and Data Sharing tabs are available on the Users page.
3. Open the Permissions tab and verify the available permission sets.

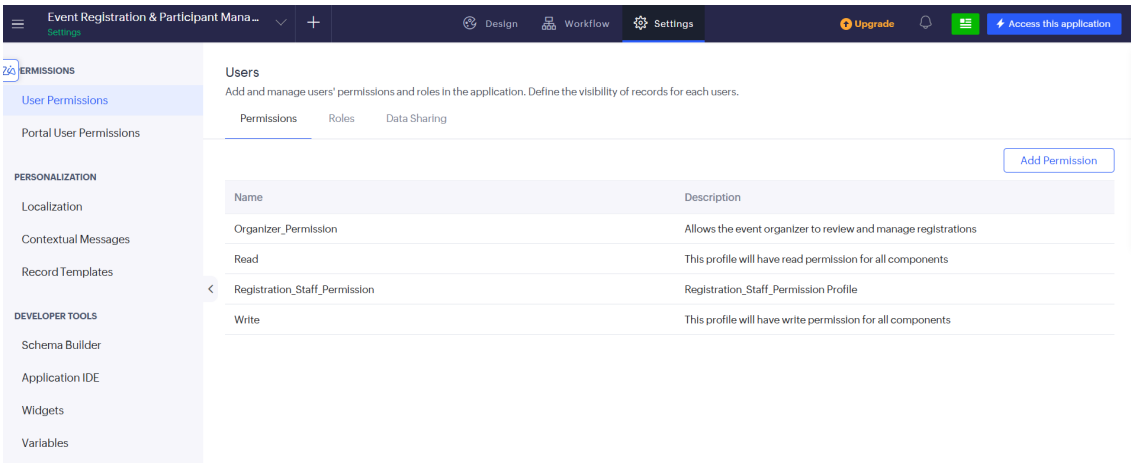



*Figure 8.1: User Permissions tab showing Organizer_Permission, Registration_Staff_Permission, Read and Write.*

## Procedure (Continued)

4. Open the **Roles** tab. The role tree shown in the application is CEO → Event_Organizer → Registration_Staff.
5. Verify that Event_Organizer is nested under CEO and Registration_Staff is nested under Event_Organizer.

This structure represents the reporting hierarchy for the application.

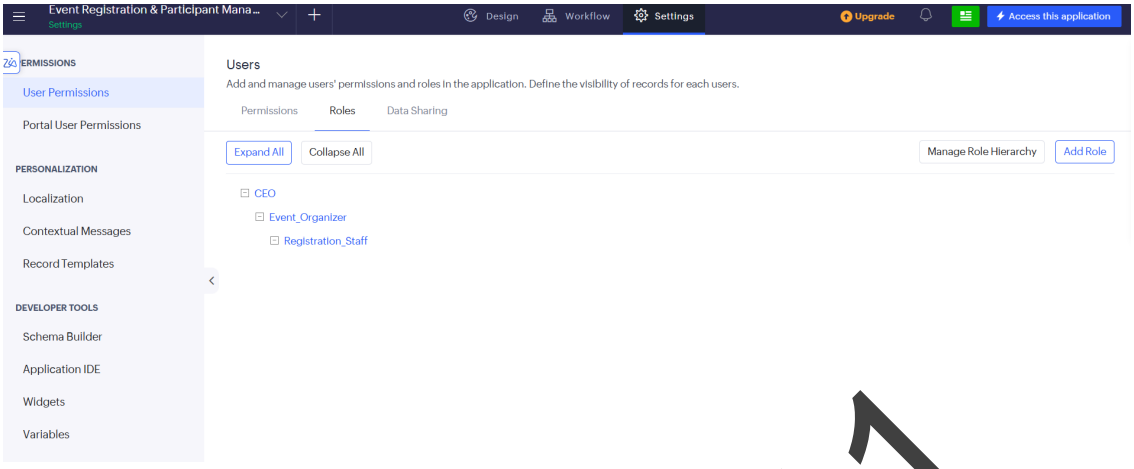



*Figure 8.2: Roles tab displaying the CEO → Event_Organizer → Registration_Staff hierarchy.*

### Role hierarchy interpretation

The CEO occupies the top level, Event_Organizer is the middle level, and Registration_Staff is the subordinate level. The hierarchy helps determine record visibility for users assigned to these roles, together with their permission sets.

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## Procedure (Continued): Data Sharing

6. Open the **Data Sharing** tab and choose **New Share Rule**.
7. In the New Share Rule panel, select **Event_Registration_Form** in the Form field.
8. Enter a descriptive rule name, for example **Registration_Staff_ReadOnly_Sharing**.
9. Choose the appropriate source and destination roles in the Records shared from and Records shared to fields. The screenshot below records the current visible selections in the rule configuration panel.
10. Set **Level of access** to **Read Only** when the receiving role should view the shared records without editing them.
11. Review the rule settings and click **Save** to add the rule.

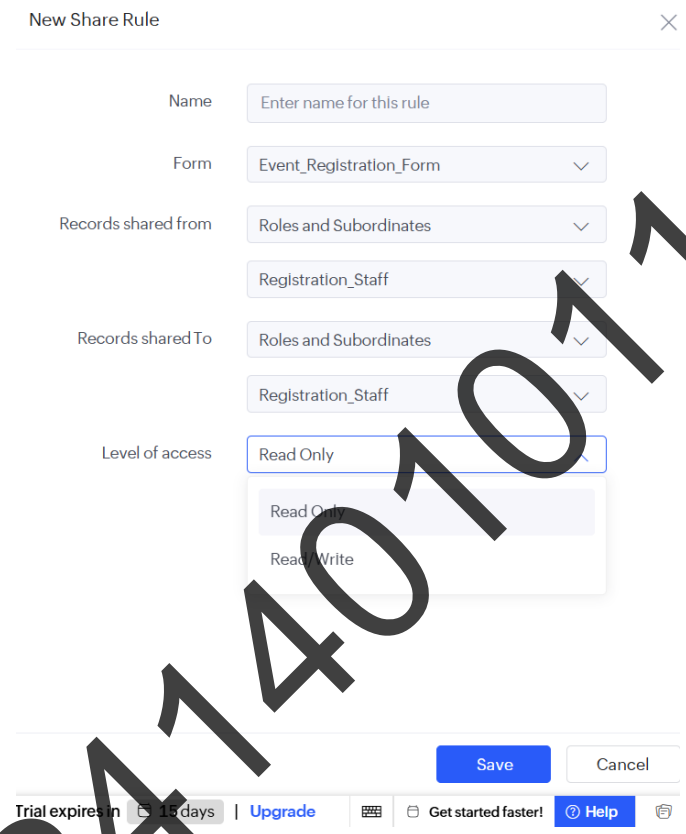



*Figure 8.3: New Share Rule panel for Event_Registration_Form with Read Only access selected.*

**Note:** The supplied screenshot shows the rule-creation panel before saving. Verify the source and destination selections match the intended sharing policy before saving; the screenshot alone does not confirm that a rule has been saved.

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## RBAC Working

| Level | Role | Intended Function |
|---|---|---|
| 1 (Top) | CEO | Overall authority and oversight within the application. |
| 2 | Event_Organizer | Coordinates event-level registration management. |
| 3 | Registration_Staff | Handles registration-related work under the organizer role. |

The permission sets determine which components and actions users may access. The role hierarchy establishes reporting relationships, while data-sharing rules can extend record visibility where required. A Read Only rule allows recipients to view shared records without changing them.

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## Result

• The custom permission sets Organizer_Permission and Registration_Staff_Permission are listed in the Permissions tab.
• The role hierarchy CEO → Event_Organizer → Registration_Staff is displayed in the Roles tab.
• The New Share Rule configuration panel is opened for Event_Registration_Form and Read Only is selected as the access level.
• The configuration should be saved after verifying the source and destination role selections.